

Two-Stage Cascade Interactive PET/CT Lesion Segmentation with nnU-Net

Yibo Wang^[0009–0009–8934–9754]

Nanjing University, China
ybwang@smail.nju.edu.cn

Abstract. We present a two-stage cascade interactive lesion segmentation pipeline for the autoPET V challenge on whole-body multi-tracer PET/CT. An initial 2-channel PlainConvUNet predicts a baseline segmentation from CT and PET without click guidance. A cascade PlainConvUNet then refines the prediction under simulated clinician scribbles (foreground or background Gaussian click maps), taking the previous-stage segmentation as an additional input channel (5-channel cascade). Both stages are trained from scratch with nnU-Net v2 on the official autoPET training cohort ($n=1614$; 1014 FDG + 600 PSMA). At inference we disable test-time mirroring, select a single full-data checkpoint (no cross-validation ensemble), and apply a connected-component size filter (35 mm^3) to suppress small false positives. The design targets a strong zero-click baseline while remaining responsive to sparse corrective interactions.

Keywords: autoPET challenge · PET/CT segmentation · human-AI interaction · nnU-Net · cascade.

1 Introduction

Malignant tumors are one of the leading causes of death in humans, and oncological imaging helps support diagnosis, staging, evaluation of therapy response, and disease monitoring. In oncological imaging, Positron Emission Tomography combined with Computed Tomography (PET/CT) is necessary and widely used. In current clinical practice, radiologists assess tumor burden by visually inspecting PET/CT scans and identifying changes in the size and distribution of the lesion using standardized criteria. In this manual process, only a limited subset of lesions is typically evaluated using simplified one-dimensional measurements. Consequently, a large portion of the rich quantitative and spatial information available in PET imaging remains underutilized. Furthermore, manual assessment is subject to inter-observer variability and may not fully capture complex disease patterns.

Recent advances in deep learning have significantly improved the segmentation of whole-body PET/CT lesions; however, some issues remain. These include limited robustness to variations between scanners, tracers, and imaging centers, difficulties in distinguishing physiological uptake from pathological lesions, and

degraded performance in low-contrast or clinically complex cases. These issues mean that expert review and correction are still indispensable.

To better reflect clinical practice, the autoPET V challenge [2] focuses on interactive lesion segmentation in whole-body PET/CT imaging. Unlike conventional fully automatic segmentation tasks, interactive segmentation considers an iterative workflow in which clinicians provide sparse corrective feedback to refine model predictions, providing a comprehensive assessment of segmentation precision, interaction efficiency, and robustness to diverse clinical scenarios.

The autoPET V challenge [2] is hosted at MICCAI 2026 [4] in collaboration with the European Society of Radiology (ESR) under the initiative “AI-based assessment of PET imaging for oncology” [1] and is supported by the European Society for Hybrid, Molecular and Translational Imaging (ESHI) [3].

Our contribution is a practical two-stage nnU-Net cascade tailored to this setting: a click-free initial model for a robust starting prediction, followed by a click-conditioned cascade refiner, with lightweight connected-component post-processing. Algorithm details are summarized in Table 2.

2 Methods

We describe the submitted algorithm below; a compact summary is given in Table 2.

2.1 Data

Training data We use the official autoPET V Task 1 training cohort comprising $n=1614$ whole-body PET/CT studies with lesion annotations: 1014 FDG and 600 PSMA examinations [5, 9, 2]. All labeled training cases are used for model fitting (full-data training; see below). Click/scribble channels for the cascade stage are generated with the challenge interactive simulation pipeline (Gaussian foreground/background heatmaps).

Validation data We do not hold out a classical cross-validation fold for model selection. Instead, models are trained on the full training set, and checkpoints are selected by the lowest training loss (nnU-Net full-data training protocol). No separate external validation cohort beyond the challenge training data was used for hyper-parameter search.

2.2 Data pre-processing

Pre-processing follows the default nnU-Net v2 preprocessor [7, 8]: CT intensity normalization, PET (and click-map) z-score normalization, and resampling to configuration-specific spacings. The cascade low-resolution configuration uses coarser spacing ($\approx [5.26, 3.57, 3.57]$ mm), while full-resolution and cascade full-resolution use $\approx [3.0, 2.04, 2.04]$ mm. All configurations share a 3D patch size of $112 \times 160 \times 128$.

2.3 Algorithm/model

Overview Inference is organized as an interactive loop. At iteration 0 (empty clicks), a 2-channel PlainConvUNet predicts lesions from CT and PET only. When corrective scribbles are available, a cascade PlainConvUNet refines the segmentation using CT, PET, foreground/background click maps, and the previous-stage segmentation as an extra input channel (effective 5-channel input), trained under the nnU-Net cascade full-resolution configuration.

Initial model (click-free) Dataset modality layout: CT + PET (2 channels). Architecture: 3D PlainConvUNet from nnU-Net plans (not Residual Encoder). We train for up to 3000 epochs (83 iterations/epoch, batch size 6) with SGD and polynomial learning-rate decay (initial LR 0.01), compound Dice + cross-entropy loss. The deployed initial weights are the full-data checkpoint with the lowest training loss.

Cascade refiner (click-conditioned) We first train a low-resolution PlainConvUNet from scratch on the four modality channels (CT, PET, and foreground/background clicks), then train the cascade full-resolution network with previous-stage segmentations as additional input. Both stages use the same patch size, batch size, optimizer/scheduler, and Dice+CE loss; the cascade stage is trained for up to 3000 epochs. The submitted cascade weights are likewise selected by the lowest training loss. At inference, the previous-stage map fed to the cascade is produced by the 2-channel full-resolution initial model (rather than re-running the low-resolution stage at test time), while the cascade network itself remains the model trained in the standard low-resolution→cascade curriculum.

Interactive feedback Simulated scribbles follow the challenge centerline-based interaction strategy. Foreground/background points are converted to Gaussian heatmaps and concatenated as click channels for cascade refinement across interaction rounds.

2.4 Data post-processing

After each prediction we apply a connected-component filter that removes components smaller than 35 mm^3 . This targets small false-positive blobs that disproportionately hurt lesion-level detection metrics, without changing the network thresholding itself. No PET-SUV thresholding and no test-time mirroring are used.

2.5 Training and test parameters

Training Framework: nnU-Net v2 [7]. Optimizer/scheduler: SGD with Nesterov momentum and polynomial LR schedule (nnU-Net defaults; initial LR 0.01).

Batch size: 6; patch: $112 \times 160 \times 128$; 83 training iterations per epoch. Data augmentation: nnU-Net v2 default DA (Gaussian blur, contrast, gamma, mirroring, rotation, scaling, low-resolution simulation; elastic deformation disabled by default in nnU-Net v2). All networks (initial, low-resolution, and cascade) are randomly initialized and trained from scratch on autoPET data. Hardware: $3 \times$ NVIDIA RTX 3090.

Testing / inference Test-time augmentation: none. Ensembling: none (single full-data model; no multi-fold averaging). Sliding-window inference follows nnU-Net defaults. Post-processing: connected-component minimum volume 35 mm^3 .

2.6 Github repository

<https://github.com/yeebowang/autoPET-V-submission>.

3 Results

We report an official fold-0 hold-out evaluation using the challenge-provided split ($n_{\text{train}}=1291$, $n_{\text{val}}=323$). Models are trained on the fold-0 training split only (no validation step during training; checkpoint = lowest training loss), following the same recipe as the submission: 2-channel full-resolution initializer (3000 epochs) $\rightarrow$ 4-channel low-resolution stage (2000 epochs) $\rightarrow$ 5-channel cascade full-resolution refiner (3000 epochs). Interactive evaluation uses centerline-simulated scribbles for 10 iterations (iterations 0–9), without test-time augmentation, with the cascade previous-stage map taken from the 2-channel full-resolution prediction. We compare inference without connected-component post-filtering versus applying a 35 mm^3 minimum-volume filter (as in the submission).

Table 1 reports mean Dice, false-positive volume (FPV), and false-negative volume (FNV) at each iteration, together with AUC-Dice / AUC-FPV / AUC-FNV obtained by trapezoidal integration over iteration indices. The table layout follows the per-interaction Dice/FPV/FNV comparison style of prior autoPET interactive reports [6], contrasting the two post-processing settings rather than alternative training variants.

Both settings exhibit a stable zero-click baseline and progressive Dice gains as simulated scribbles accumulate, while FPV and FNV decrease over the interactive loop. At iteration 0, mean Dice is essentially identical (0.526 for both), indicating that the connected-component filter has little effect when the cascade has not yet received corrective clicks. With the first few interactions, the 35 mm^3 filter yields a small but consistent Dice advantage (e.g., 0.642 vs. 0.636 at iteration 1; 0.683 vs. 0.679 at iteration 2). In later iterations the Dice curves largely converge (0.764 at iteration 9 for both), while AUC-Dice remains slightly higher with the filter (6.477 vs. 6.471). Across iterations, FPV falls from ~ 9.5 to $\sim 7.5 \text{ ml}$ for both settings, and is slightly lower with the 35 mm^3 filter after the first click; FNV remains comparable (slightly higher with the filter, as tiny true-positive components may be removed). Overall, these results support keeping the

35 mm³ component filter in the submitted inference recipe, without sacrificing the late-iteration Dice ceiling.

Table 1. Average fold-0 interactive validation results across iterations 0–9 ($n=323$). Under each metric, columns compare inference without connected-component post-filtering (*none*) versus a 35 mm³ minimum-volume filter. Dice: mean over all cases. FPV/FNV: official autoPET volumes (ml); FPV over all cases, FNV over non-empty ground truth only ($n=204$). AUC rows are trapezoidal integrals over iteration indices, averaged over cases. $\uparrow/\downarrow$: higher/lower is better; bold marks the better value in each pair.

Iter	Dice $\uparrow$		FPV $\downarrow$		FNV $\downarrow$	
	none	35 mm ³	none	35 mm ³	none	35 mm ³
0	0.526	0.526	9.527	9.527	6.553	6.553
1	0.636	0.642	9.062	9.058	6.288	6.385
2	0.679	0.683	8.649	8.645	5.936	6.054
3	0.727	0.727	8.432	8.429	5.846	5.957
4	0.750	0.748	8.227	8.220	5.815	5.930
5	0.757	0.757	8.056	8.049	5.825	5.940
6	0.757	0.754	7.889	7.881	5.807	5.920
7	0.757	0.757	7.715	7.707	5.802	5.918
8	0.763	0.764	7.615	7.608	5.803	5.919
9	0.764	0.764	7.511	7.504	5.807	5.920
AUC	6.471	6.477	74.16	74.11	53.30	54.26

4 Discussion

Separating a click-free initial predictor from a click-conditioned cascade matches the challenge workflow: iteration 0 must already be clinically useful, while later iterations should absorb sparse corrections. Using previous-stage segmentations inside the cascade provides a strong spatial prior; replacing the low-resolution previous map by the stronger 2-channel full-resolution initial prediction at test time is a practical refinement of the standard cascade hand-off. The 35 mm³ component filter is a simple, tracer-agnostic regularizer against tiny false positives. Limitations include full-data training without classical cross-validation model selection, disabled test-time augmentation (speed/robustness trade-off), and reliance on simulated rather than clinician scribbles during development.

5 Conclusion

We proposed a two-stage PlainConvUNet cascade for interactive whole-body PET/CT lesion segmentation in autoPET V: a 2-channel click-free initializer

and a 5-channel click-aware cascade refiner, trained from scratch with nnU-Net v2, without test-time augmentation or ensembling, and with connected-component post-filtering. This yields a compact, reproducible submission aligned with interactive clinical use.

Table 2. Algorithm details

Team name	algorithm name	data pre-processing	data post-processing	training data augmentation
YiboWang	autoPETV++	nnUNetv2 preprocessor (CT/PET intensity normalization, resampling)	Connected-component filter (min. volume 35 mm ³)	nnUNetv2 default DA (Gaussian blur, contrast, gamma, mirroring, rotation, scaling, low-resolution simulation;)
test time augmentation	ensembling	standardized framework	network architecture	loss
None	None (single model: lowest training loss)	nnUNetv2 (3D)	PlainConvUNet (3D), two-stage: initial (no click) + cascade	DSC + CE
training data	data/model dimensionality and size	use of pre-trained models	GPU hardware for training	
1014 FDG + 600 PSMA PET/CT (autoPET V official, $n=1614$)	$112 \times 160 \times 128$; batch size 6	None (all trained from scratch, random initialization)	$3 \times$ NVIDIA RTX 3090	

Disclosure of Interests. The author has no competing interests to declare that are relevant to the content of this article.

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